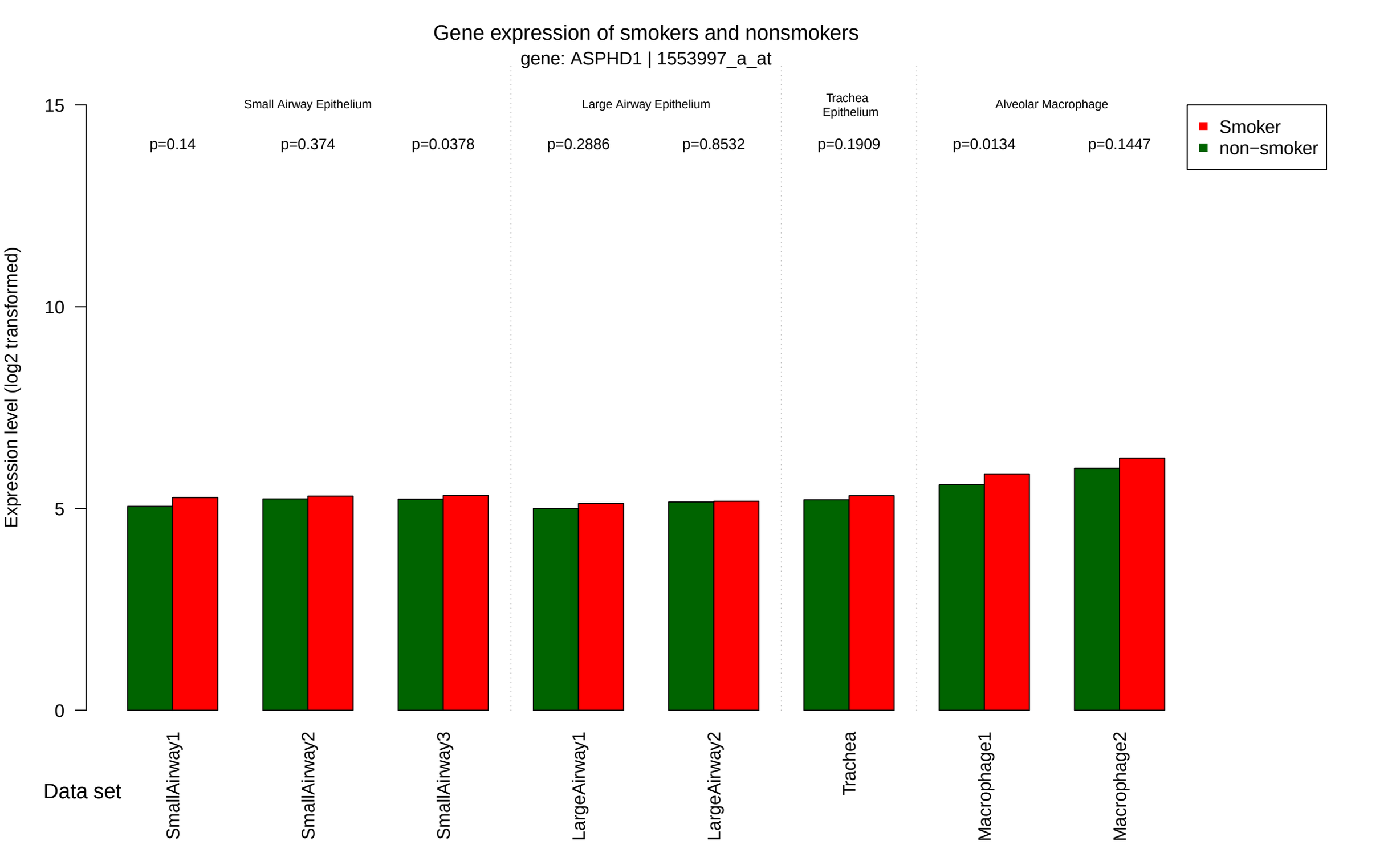
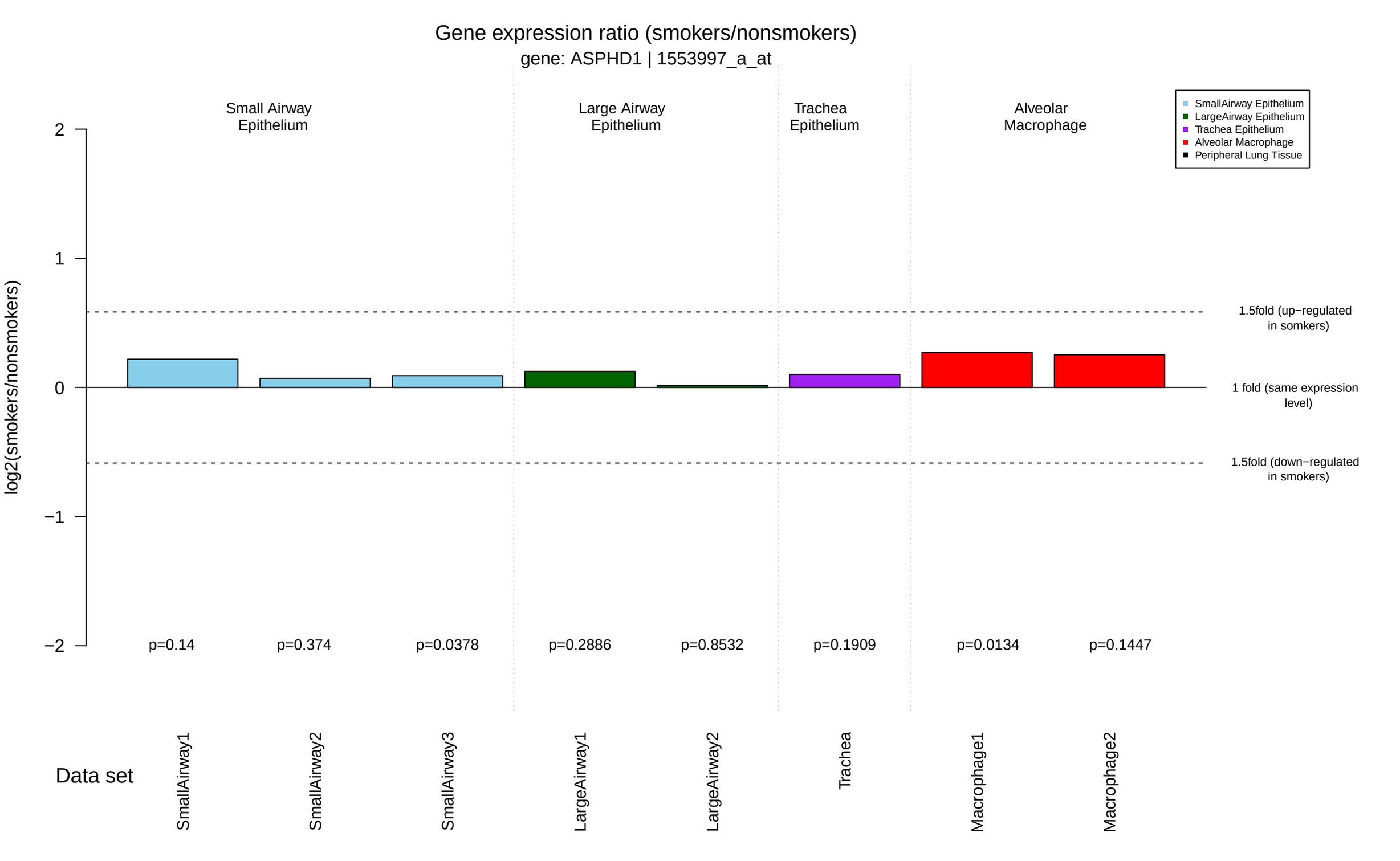
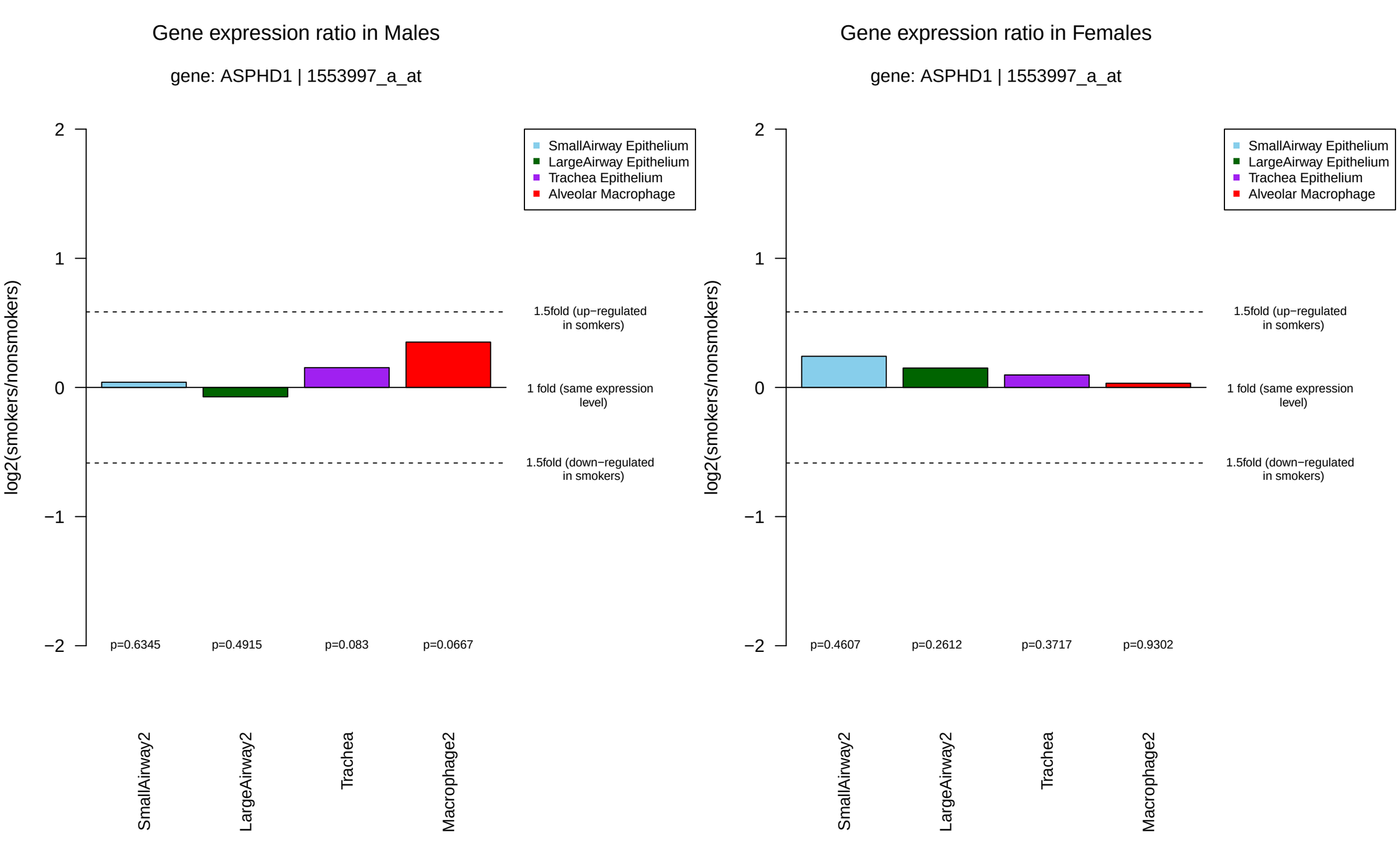
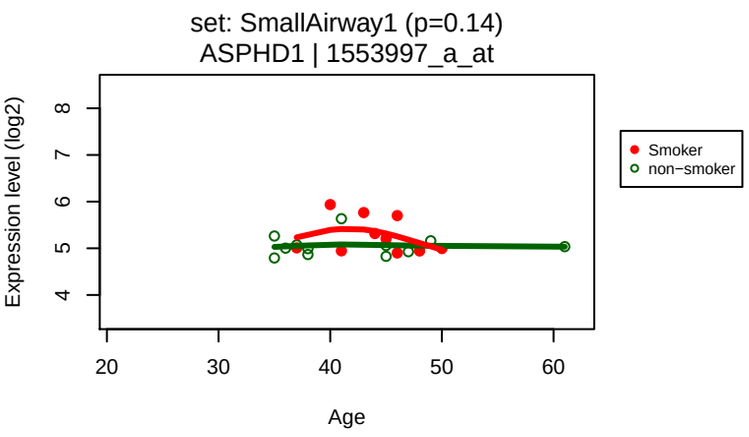
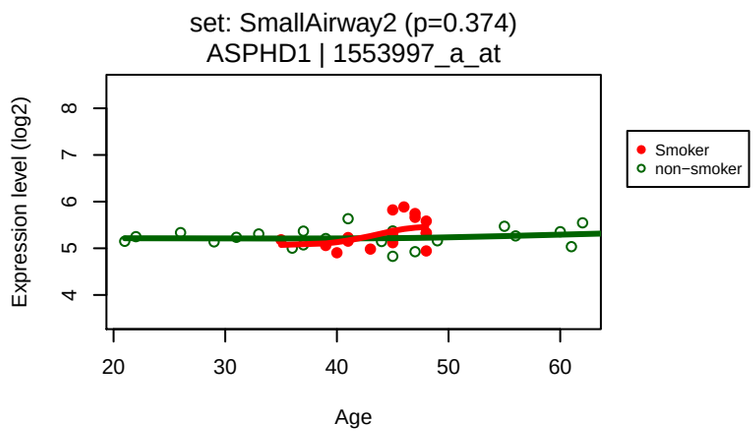
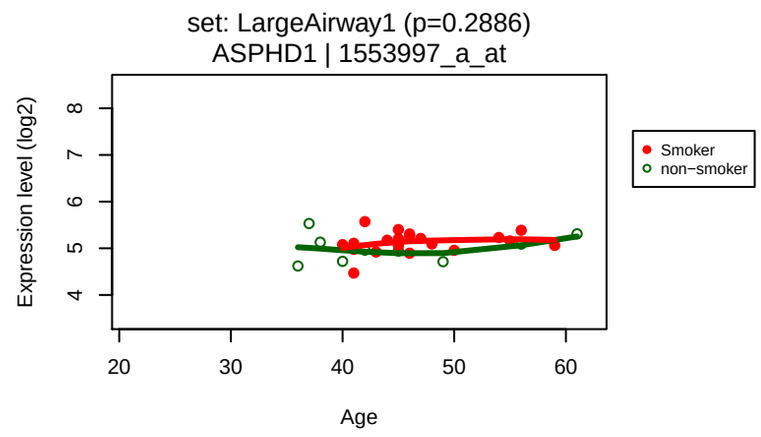

Gene expression of smokers and nonsmokers
gene: ASPHD1 | 1553997_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.14
p=0.374
p=0.0378
p=0.2886
p=0.8532
p=0.1909
p=0.0134
p=0.1447
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

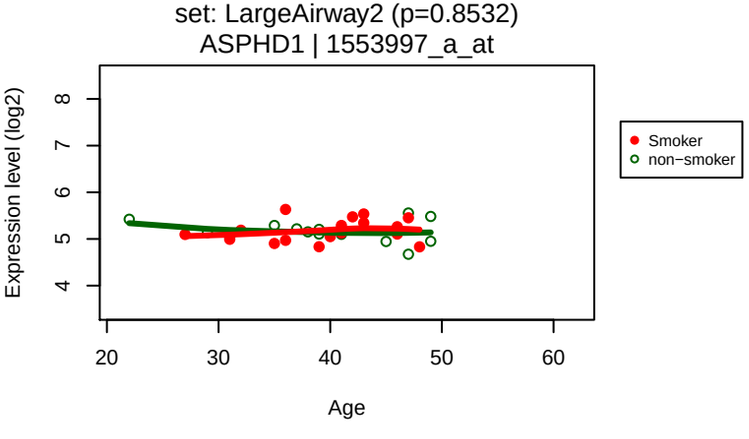

Gene expression ratio (smokers/nonsmokers)
gene: ASPHD1 | 1553997_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.14
p=0.374
p=0.0378
p=0.2886
p=0.8532
p=0.1909
p=0.0134
p=0.1447
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

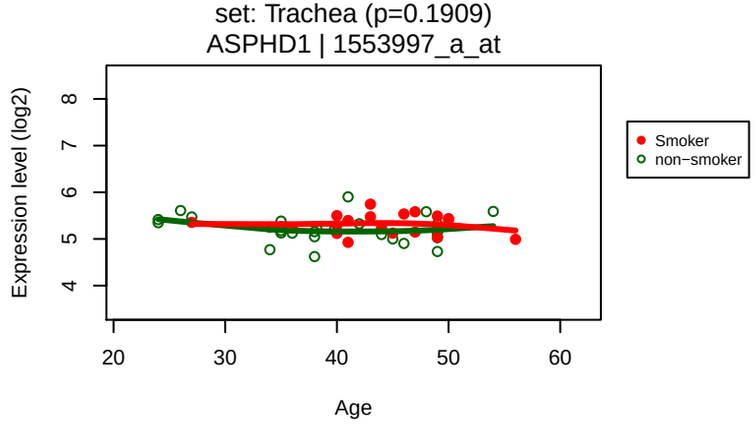

Gene expression ratio in Males
gene: ASPHD1 | 1553997_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6345
p=0.4915
p=0.083
p=0.0667
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASPHD1 | 1553997_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4607
p=0.2612
p=0.3717
p=0.9302
SmallAirway2
LargeAirway2
Trachea
Macrophage2

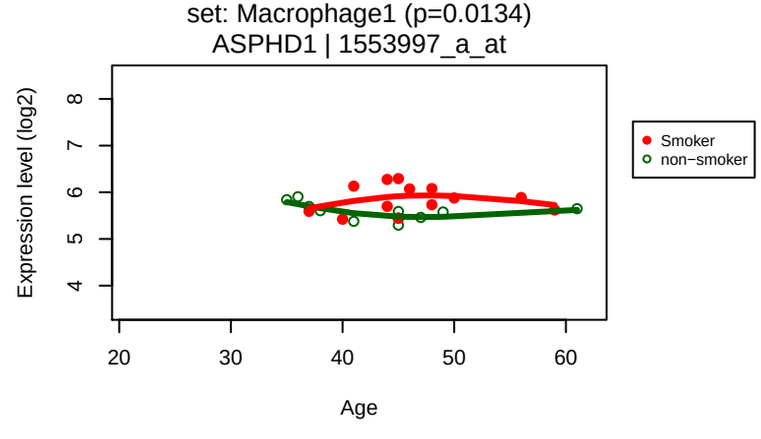

set: SmallAirway1 (p=0.14)
ASPHD1 | 1553997_a_at
Expression level (log2)
Age
Smoker
non−smoker

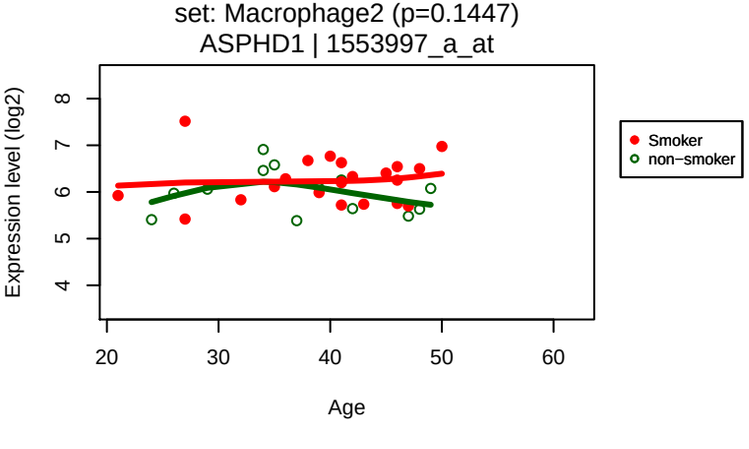

set: SmallAirway2 (p=0.374)
ASPHD1 | 1553997_a_at
Expression level (log2)
Age
Smoker
non−smoker

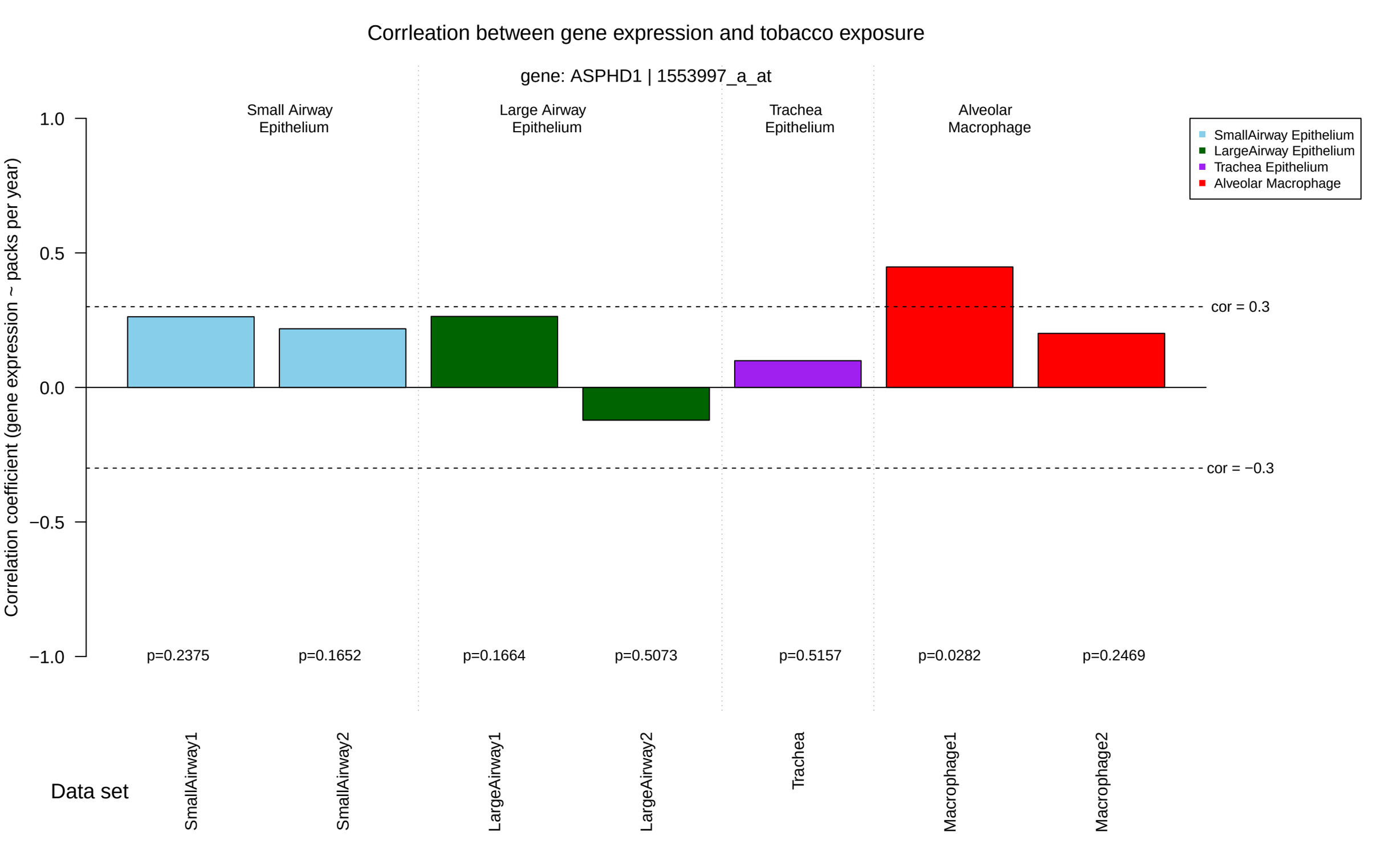

set: LargeAirway1 (p=0.2886)
ASPHD1 | 1553997_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8532)
ASPHD1 | 1553997_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1909)
ASPHD1 | 1553997_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0134)
ASPHD1 | 1553997_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1447)
ASPHD1 | 1553997_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASPHD1 | 1553997_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2375
p=0.1652
p=0.1664
p=0.5073
p=0.5157
p=0.0282
p=0.2469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2